

thuannguyenhw2

October 1, 2026

```
[2]: import pandas as pd
import re
from collections import defaultdict, Counter

df = pd.read_csv("MOVIE_REVIEWS.csv")
df.head()
```

```
[2]:
```

	Review	Rating
0	Totally waste of time and money worst movie to...	1
1	such an awful movie I have watched in a while	1
2	Worst shark film ever and did not think sharks...	1
3	The plot per se was stupid	2
4	I am very impressed how a mouth to mouth under...	2

```
[3]: def clean_text(text):
    text = text.lower()
    text = re.sub(r"[^a-z\s]", " ", text)
    text = re.sub(r"\s+", " ", text).strip()
    return text

df["Cleaned"] = df["Review"].apply(clean_text)
df["Tokens"] = df["Cleaned"].apply(lambda x: x.split())

df.head()
```

```
[3]:
```

	Review	Rating	\
0	Totally waste of time and money worst movie to...	1	
1	such an awful movie I have watched in a while	1	
2	Worst shark film ever and did not think sharks...	1	
3	The plot per se was stupid	2	
4	I am very impressed how a mouth to mouth under...	2	

	Cleaned	\
0	totally waste of time and money worst movie to...	
1	such an awful movie i have watched in a while	
2	worst shark film ever and did not think sharks...	
3	the plot per se was stupid	

```
4 i am very impressed how a mouth to mouth under...
```

Tokens

```
0 [totally, waste, of, time, and, money, worst, ...
1 [such, an, awful, movie, i, have, watched, in,...
2 [worst, shark, film, ever, and, did, not, thin...
3 [the, plot, per, se, was, stupid]
4 [i, am, very, impressed, how, a, mouth, to, mo...
```

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[5]: class_word_counts = defaultdict(Counter)
      class_doc_counts = Counter()

      for _, row in df.iterrows():
          rating = row["Rating"]
          class_doc_counts[rating] += 1
          class_word_counts[rating].update(row["Tokens"])
```

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[6]: total_docs = len(df)
      priors = {c: class_doc_counts[c] / total_docs for c in class_doc_counts}

      priors
```

```
[6]: {1: 0.14, 2: 0.2, 3: 0.18, 4: 0.24, 5: 0.24}
```

```
[8]: # Build global vocabulary from all classes
      global_vocab = set()
      for c in class_word_counts:
          global_vocab.update(class_word_counts[c].keys())

      # Test document
      test_doc = "watched this beautiful and adventure filled film just now and it_
                  ↪was the best film ever"
      test_clean = clean_text(test_doc)
      test_tokens = test_clean.split()

      # Drop unknown words
      updated_test_tokens = [w for w in test_tokens if w in global_vocab]
      updated_test_tokens
```

```
[8]: ['watched',
      'this',
      'beautiful',
      'and',
      'adventure',
      'filled',
      'film',
      'just',
```

```
'and',  
'it',  
'was',  
'the',  
'film',  
'ever']
```

```
[9]: def likelihood(word, rating):  
    vocab_size = len(global_vocab)  
    word_count = class_word_counts[rating][word]  
    total_words = sum(class_word_counts[rating].values())  
    return (word_count + 1) / (total_words + vocab_size)
```

```
[10]: scores = {}  
  
for c in class_word_counts:  
    score = priors[c]  
    for w in updated_test_tokens:  
        score *= likelihood(w, c)  
    scores[c] = score  
  
scores
```

```
[10]: {1: 3.8857805861880497e-32,  
      2: 3.7263853219749853e-31,  
      3: 5.3862533431046995e-33,  
      4: 8.774300396252768e-32,  
      5: 1.2747693903788626e-30}
```

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[11]: predicted_class = max(scores, key=scores.get)  
      predicted_class
```

```
[11]: 5
```

```
[12]: print("1. Prior probabilities:")  
      print(priors, "\n")  
  
      print("2. Updated words after dropping unknown words:")  
      print(updated_test_tokens, "\n")  
  
      print("3. Likelihood probabilities for each class:")  
      for c in class_word_counts:  
          print(f"Class {c}:")  
          for w in updated_test_tokens:  
              print(f" P({w}|{c}) = {likelihood(w, c)}")  
          print()
```

```

print("4. Final scores for all classes:")
print(scores, "\n")

print("5. Predicted class label:")
print(predicted_class)

```

1. Prior probabilities:

{1: 0.14, 2: 0.2, 3: 0.18, 4: 0.24, 5: 0.24}

2. Updated words after dropping unknown words:

['watched', 'this', 'beautiful', 'and', 'adventure', 'filled', 'film', 'just',
'and', 'it', 'was', 'the', 'film', 'ever']

3. Likelihood probabilities for each class:

Class 1:

$P(\text{watched}|1) = 0.00625$
 $P(\text{this}|1) = 0.0125$
 $P(\text{beautiful}|1) = 0.003125$
 $P(\text{and}|1) = 0.0125$
 $P(\text{adventure}|1) = 0.003125$
 $P(\text{filled}|1) = 0.003125$
 $P(\text{film}|1) = 0.0125$
 $P(\text{just}|1) = 0.003125$
 $P(\text{and}|1) = 0.0125$
 $P(\text{it}|1) = 0.003125$
 $P(\text{was}|1) = 0.003125$
 $P(\text{the}|1) = 0.0125$
 $P(\text{film}|1) = 0.0125$
 $P(\text{ever}|1) = 0.0125$

Class 2:

$P(\text{watched}|2) = 0.002583979328165375$
 $P(\text{this}|2) = 0.012919896640826873$
 $P(\text{beautiful}|2) = 0.002583979328165375$
 $P(\text{and}|2) = 0.023255813953488372$
 $P(\text{adventure}|2) = 0.002583979328165375$
 $P(\text{filled}|2) = 0.002583979328165375$
 $P(\text{film}|2) = 0.007751937984496124$
 $P(\text{just}|2) = 0.007751937984496124$
 $P(\text{and}|2) = 0.023255813953488372$
 $P(\text{it}|2) = 0.0103359173126615$
 $P(\text{was}|2) = 0.007751937984496124$
 $P(\text{the}|2) = 0.031007751937984496$
 $P(\text{film}|2) = 0.007751937984496124$
 $P(\text{ever}|2) = 0.00516795865633075$

Class 3:

$P(\text{watched}|3) = 0.003194888178913738$
 $P(\text{this}|3) = 0.012779552715654952$
 $P(\text{beautiful}|3) = 0.003194888178913738$
 $P(\text{and}|3) = 0.009584664536741214$
 $P(\text{adventure}|3) = 0.003194888178913738$
 $P(\text{filled}|3) = 0.003194888178913738$
 $P(\text{film}|3) = 0.006389776357827476$
 $P(\text{just}|3) = 0.003194888178913738$
 $P(\text{and}|3) = 0.009584664536741214$
 $P(\text{it}|3) = 0.006389776357827476$
 $P(\text{was}|3) = 0.009584664536741214$
 $P(\text{the}|3) = 0.009584664536741214$
 $P(\text{film}|3) = 0.006389776357827476$
 $P(\text{ever}|3) = 0.003194888178913738$

Class 4:

$P(\text{watched}|4) = 0.0028089887640449437$
 $P(\text{this}|4) = 0.014044943820224719$
 $P(\text{beautiful}|4) = 0.0056179775280898875$
 $P(\text{and}|4) = 0.014044943820224719$
 $P(\text{adventure}|4) = 0.0028089887640449437$
 $P(\text{filled}|4) = 0.0028089887640449437$
 $P(\text{film}|4) = 0.0056179775280898875$
 $P(\text{just}|4) = 0.0056179775280898875$
 $P(\text{and}|4) = 0.014044943820224719$
 $P(\text{it}|4) = 0.011235955056179775$
 $P(\text{was}|4) = 0.011235955056179775$
 $P(\text{the}|4) = 0.016853932584269662$
 $P(\text{film}|4) = 0.0056179775280898875$
 $P(\text{ever}|4) = 0.0028089887640449437$

Class 5:

$P(\text{watched}|5) = 0.0025575447570332483$
 $P(\text{this}|5) = 0.015345268542199489$
 $P(\text{beautiful}|5) = 0.0076726342710997444$
 $P(\text{and}|5) = 0.01278772378516624$
 $P(\text{adventure}|5) = 0.005115089514066497$
 $P(\text{filled}|5) = 0.005115089514066497$
 $P(\text{film}|5) = 0.0076726342710997444$
 $P(\text{just}|5) = 0.005115089514066497$
 $P(\text{and}|5) = 0.01278772378516624$
 $P(\text{it}|5) = 0.010230179028132993$
 $P(\text{was}|5) = 0.010230179028132993$
 $P(\text{the}|5) = 0.02557544757033248$
 $P(\text{film}|5) = 0.0076726342710997444$
 $P(\text{ever}|5) = 0.005115089514066497$

4. Final scores for all classes:

```
{1: 3.8857805861880497e-32, 2: 3.7263853219749853e-31, 3:  
5.3862533431046995e-33, 4: 8.774300396252768e-32, 5: 1.2747693903788626e-30}
```

5. Predicted class label:

5

[]: